

Tumor Segmentation in Brain MR Images Using Res-UNet with Multimodal Fusion

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Abstract—Early diagnosis of brain tumors presents significant challenges. Single-modality imaging often suffers from limitations, such as incomplete structural representation and a lack of complementary information, which reduce segmentation accuracy and complicate the identification of complex tumor scenarios. Additionally, as network depth increases, performance degradation becomes a notable issue. This research presents a multimodal fusion approach that efficiently combines complementary information from various MRI modalities to overcome these challenges. This approach mitigates the limitations of single-modality data, providing the model with richer and more comprehensive features, thereby enhancing the characterization of tumor regions. The Res-UNet model is employed to merge the benefits of the U-Net architecture for biomedical image segmentation with the strengths of residual modules. This combination effectively tackles the degradation issue linked to increased network depth, markedly enhancing segmentation accuracy, especially in defining tumor boundaries and distinguishing different tumor regions. The experimental outcomes on public datasets reveal that the proposed model surpasses existing ones in all evaluation metrics. It can preserve lesion details and precisely segment tumor boundaries. This advancement offers substantial support for clinical brain tumor diagnosis and contributes to the fields of medical imaging, bioinformatics, and related disciplines.

Keywords—Brain Tumor Segmentation, Multimodal Fusion, Res-UNet, MRI

I. INTRODUCTION

Brain tumors are a major health threat, and their early detection and precise treatment have long been central concerns in medical research. With the continuous advancement of medical imaging technologies, Magnetic Resonance Imaging (MRI) has become a crucial instrument for the diagnosis of brain tumors. Nonetheless, the early identification of these tumors continues to pose a significant challenge. Early symptoms are often non-specific and can easily be overlooked or misdiagnosed, leading to missed opportunities for timely intervention. Furthermore, brain tumors vary greatly in location and size, which leads to a

wide spectrum of symptoms, complicating the diagnosis. While MRI offers detailed images of soft brain tissues, manually segmenting tumor regions is still labor-intensive and time-consuming. The intricate nature of brain structures, the unclear boundaries of tumors, and individual anatomical differences contribute to inconsistencies in manual segmentation, creating considerable challenges for clinicians aiming for accurate diagnoses.

The swift advancement of deep learning technologies opens up new possibilities for tackling the challenges associated with brain tumor segmentation. Among the various deep learning architectures, U-Net and its derivatives have demonstrated considerable effectiveness in medical image segmentation. The U-Net's encoder-decoder architecture facilitates the efficient extraction of contextual information from images, leading to impressive performance in biomedical segmentation tasks. Despite this, U-Net faces limitations when applied to complex brain tumor images. For example, merely increasing the network depth can lead to issues such as vanishing gradients, which degrade model performance and segmentation accuracy. In addition, brain tumor MRI scans typically include multiple imaging modalities, each providing complementary information. Effectively integrating this multimodal data is crucial for enhancing segmentation accuracy.

To address these challenges, we propose a Res-UNet model that incorporates multimodal fusion for the segmentation of brain tumors in MRI scans. The multimodal fusion strategy combines information from different MRI sequences, enabling the model to extract richer and more comprehensive feature representations, thereby enhancing its ability to identify tumor regions accurately. Furthermore, by incorporating residual modules into the U-Net framework, the Res-UNet model mitigates the degradation issues often encountered in deeper networks. These residual connections help ensure robust performance when processing complex brain tumor images and improve segmentation accuracy.

Accurate tumor segmentation in clinical practice is essential for helping physicians understand the tumor's location, size, morphology, and its relationship with surrounding structures. This comprehensive segmentation aids in creating personalized treatment strategies, ultimately enhancing therapeutic outcomes and increasing patient survival rates.

II. RELATED WORK

Segmenting brain tumors is a crucial task in the realm of medical image analysis. Traditional approaches often rely on manual delineation by radiologists or semi-automated techniques based on image processing algorithms. These methods commonly employ threshold segmentation, edge detection, and morphological operations to identify tumor regions. However, their dependence on predefined thresholds and manual intervention limits their effectiveness, often resulting in inconsistent outcomes.

The advent of deep learning has transformed medical image analysis, including the segmentation of brain tumors. The U-Net^[1] architecture introduced by Ronneberger et al. has become the benchmark model for biomedical image segmentation tasks. U-Net employs contraction paths to gather contextual information and symmetric expansion paths to ensure precise localization. Variants like UNet++^[2] and Attention U-Net^[3] enhance segmentation accuracy by incorporating dense connectivity and attention mechanisms, respectively.

These models utilize multimodal data, including T1-weighted, T2-weighted, and FLAIR MRI sequences, to gather complementary information about the tumor, adjusting the influence of each modality according to its significance for the segmentation task. By fusing multimodal data, more accurate and robust segmentation results can be achieved, significantly enhancing segmentation performance. Moreover, models that integrate residual networks (ResNets) with the U-Net framework show significant promise in the segmentation of brain tumors. Residual blocks facilitate the training of deeper networks by mitigating the prevalent issue of vanishing^[4] gradients in deep architectures. The Res-UNet framework capitalizes on the advantages of both U-Net and ResNet, enhancing segmentation accuracy by effectively capturing both local and global features.

Typically, the assessment of brain tumor segmentation models involves metrics such as the Dice similarity coefficient (DSC), sensitivity, specificity, and Hausdorff distance. The DSC is a commonly used measure that quantifies the overlap between the predicted segmentation masks and the ground truth. Sensitivity and specificity offer insights into the model's ability to accurately differentiate between positive and negative cases. The Hausdorff distance^[5] measures the spatial discrepancy between the predicted and actual segmentation boundaries.

In conclusion, the advancement of deep learning models, especially those that incorporate residual networks and multimodal fusion, has greatly propelled the field of brain tumor segmentation. These models hold the promise of delivering more accurate, efficient, and automated segmentation, ultimately enhancing patient diagnosis and treatment planning.

III. METHOD

In the brain tumor segmentation task, our primary focus is on accurately distinguishing tumor regions from normal brain tissues, particularly addressing challenges such as the diversity of tumor morphology and location, as well as the effective fusion of multimodal information from MRI images. The model we propose is based on the U-Net architecture, utilizing its advantages in medical image segmentation and improving performance by integrating residual modules. Figure 1 illustrates the overall network structure.

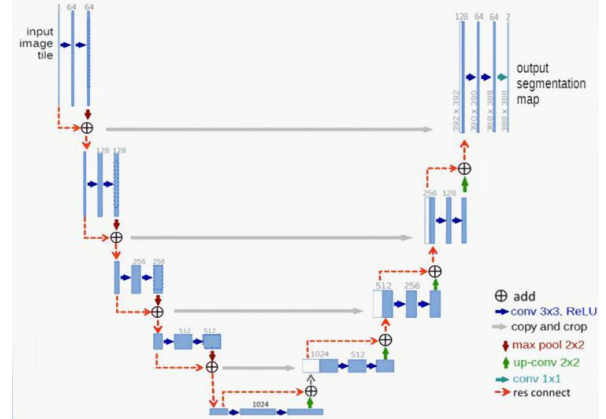


Figure 1. The proposed Res-UNet architecture.

A. Input layer

Before being input into the network, the images undergo modal sequence processing. Since the four modal sequences of brain tumors (T1, T2, FLAIR, T1ce) highlight different aspects of image information, the processing stage integrates the information from these four modalities by assigning weights to each modal image. The weight values are determined through the following steps:

First, a weight value is assigned to each of the four MRI modalities (T1, T2, T1C, FLAIR). Second, the images from each modality are individually trained and tested to obtain their respective Dice coefficient values, denoted as $Dice_{T1}$, $Dice_{T2}$, $Dice_{T1C}$ and $Dice_{FLAIR}$. Third, by analyzing the segmentation outcomes for each modality, we assess the impact of each modality on the final segmentation, allowing us to adjust the corresponding weights accordingly.

$$Dice_{sum} = Dice_{T1} + Dice_{T2} + Dice_{T1C} + Dice_{FLAIR} \quad (1)$$

$$(\alpha, \beta, \gamma, \theta) = \left(\frac{Dice_{T1}}{Dice_{sum}}, \frac{Dice_{T2}}{Dice_{sum}}, \frac{Dice_{T1C}}{Dice_{sum}}, \frac{Dice_{FLAIR}}{Dice_{sum}} \right) \quad (2)$$

As illustrated in Eq. (1) and Eq. (2), the Dice coefficient values for each modality are derived through training and testing. Based on these values, the weights are determined, and the four modal images are linearly fused according to their respective weights to serve as the input to the segmentation network. This process simulates information fusion and enhances the quality of the input images.

B. Encoder-Decoder Architecture

The encoder consists of five blocks, each containing a sequence of convolutional layers, with downsampling accomplished through convolutional layers that use a stride

of 2. Following each downsampling step, the size of the feature map is reduced by half, while the number of channels is increased twofold. This process enables the gradual extraction of deep image features, capturing richer feature information by progressively increasing the channel count while simultaneously reducing the feature map size for efficient subsequent processing. Additionally, residual modules are incorporated after each convolutional layer in the encoder. These residual modules play a crucial role in feature extraction. Each residual module is made up of two $3 \times 3 \times 3$ convolutional layers along with two batch normalization (BN) layers. The output of the first BN layer is activated using the ReLU activation function. This configuration improves feature extraction and helps mitigate the vanishing gradient issue via residual connections, facilitating the construction of deeper networks that can capture more complex image features.

The decoder is made up of four blocks, where upsampling is carried out using a transposed convolutional layer with a stride of 2. After each upsampling step, the output feature map dimensions are doubled, gradually restoring the feature map to match the size of the input image. This process of transposed convolutional upsampling reconstructs the compressed feature maps from the encoder back to their original dimensions, aiding in the final output and comparison, ensuring the model accurately learns intricate details in the image. Additionally, the decoder employs a skip-connection mechanism^[6], which directly connects layers with matching feature map sizes from the downsampling (encoder) and upsampling (decoder) processes. This mechanism effectively fuses shallow and deep features: shallow features capture local details of the image, while deep features encode more abstract semantic information. By utilizing skip connections, the model optimally harnesses these features, increasing resilience to noise, reducing the gradient vanishing issues typically associated with deeper networks, and speeding up model convergence. Additionally, this approach to feature reuse minimizes the number of feature maps and decreases memory usage.

C. Residual Structure

In Res-UNet, each residual module consists of two convolutional layers paired with two normalization layers. As shown in Fig. 2, the mathematical formulation of a simple residual module, given an input, is as follows:

The first convolutional layer is defined as:

$$y_1 = W_1x + b_1, \quad (3)$$

where W_1 stands for the convolution kernel weights and b_1 indicates the bias. After batch normalization and activation function:

$$z_1 = \text{ReLU}(\text{BN}(y_1)) . \quad (4)$$

The second convolutional layer is defined as:

$$y_2 = W_2z_1 + b_2 . \quad (5)$$

The final output of the residual module is:

$$z_2 = \text{ReLU}(\text{BN}(y_2)) + x, \quad (6)$$

where the $+x$ term represents the residual connection, facilitating gradient flow during backpropagation and mitigating the gradient vanishing problem.

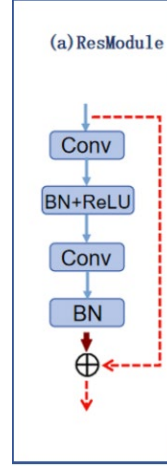


Figure 2. Residual Structure.

D. Multi-Class Focal Loss Function

In the model training stage, the multi-class focal loss function (MFLF)^{[7]-[9]} was employed to select a more suitable optimization algorithm. The formula is as follows:

$$\text{MFL}(p) = - \sum_{i=1} \alpha_i (1 - p_i)^\gamma \log(p_i), \quad (7)$$

Where p_i represents the probability of the model predicting the sample to be in the i -th category, and α_i is the category weight. Given that the number of samples across different categories can vary greatly, directly applying the cross-entropy loss function may lead the model to favor categories with a larger number of samples. By assigning different weights to different categories, we can balance this disparity, enabling the model to focus more on categories with fewer samples. The parameter γ is the focusing parameter, which determines the model's attention to hard samples. When $\gamma = 0$, the focal loss reduces to the standard cross-entropy loss function. As γ increases, the model increasingly concentrates on challenging samples, thereby improving its ability to learn the features of those samples.

IV. EXPERIMENT

To evaluate the effectiveness of our proposed approach, we compare our models against the conventional U-Net, multimodal fusion U-Net, and two-channel U-Net for segmentation. All models are retrained on the same dataset utilized in our experiments, adhering to their default parameters. Ultimately, a comprehensive analysis of the experimental results is conducted.

A. Datasets

The dataset used in this project is derived from the BRATS2015 dataset, a widely used brain tumor dataset, consisting of 274 training samples. Among these, 250 samples are for model training and the other 24 samples are kept for testing. The model is trained using randomly selected patches of dimensions $4 \times 128 \times 128 \times 128$ from the 250 training samples derived from various MRI image sequences. Normal brain tissue is labeled as category 0,

while the lesion regions are classified into four categories: necrosis (category 1), edema (category 2), non-enhancing (category 3), and enhancing tumors (category 4). The dataset is further divided into two distinct brain tumor training sets: HGG (anaplastic astrocytoma and glioblastoma multiforme) and LGG (astrocytoma and oligodendroglioma), in equal proportions. Because of the small size of the test set and the potential for low accuracy, cross-validation^{[10]-[12]} is employed to determine the appropriate proportions of the training and test sets, as shown in Table I.

TABLE I. DIVISION OF TWO DIFFERENT BRAIN TUMOR TRAINING SETS.

	HGG	LGG	Total
Training Set	200	50	250
Test Set	20	4	24
Total	220	54	274

B. Evaluation Metrics

In this experiment, we utilize specificity, sensitivity, and the Dice similarity coefficient (DSC) as metrics for evaluating the segmentation performance of the model.

The specificity is defined as:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (8)$$

The sensitivity is defined as:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (9)$$

The Dice similarity coefficient is defined as:

$$\text{DSC} = \frac{2\text{TP}}{\text{FP} + 2\text{TP} + \text{FN}} \quad (10)$$

The positive predictive value (PPV) is defined as:

$$\text{PPV} = \frac{\text{TP}}{\text{FP} + \text{TP}} \quad (11)$$

Where FP denotes false positives, TP represents true positives, TN indicates true negatives, and FN stands for false negatives.

C. Results

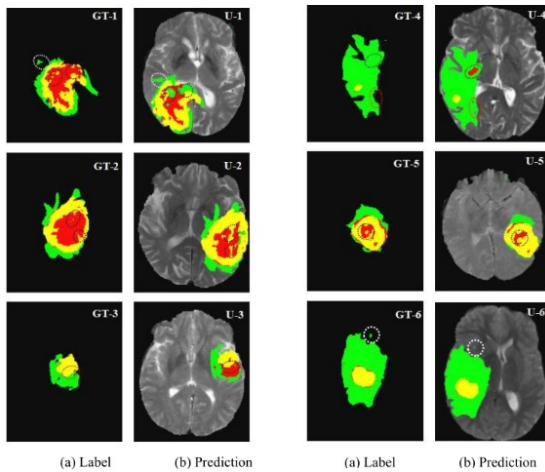


Figure 3. Segmentation Results

The segmentation results using the Res-UNet model are analyzed as follows. As illustrated in Figure 3, the green region represents edema, the blue region denotes the enhancing tumor, the red region indicates necrotic tumor, and the yellow region corresponds to the active tumor. By contrasting the entire tumor region that was manually segmented by experts with the region delineated by the model, it becomes clear that the results are highly precise.

At the beginning of the experiment, we conducted cross-validation by evenly dividing the labeled brain tumor images into five groups. During the experiment, one group was used as the validation set for testing each time, while the remaining four groups served as the training set for model training. This procedure was carried out five times. The outcomes of the cross-validation are summarized in Table II, where HD represents the Hausdorff distance, indicating the spatial separation between two sets. A lower value implies greater similarity between the two sets.

TABLE II. CROSS-VALIDATION RESULTS

Evaluation Metrics	Dice/%	PPV/%	Sensitivity/%	HD
1	91.38	88.89	88.24	2.1718
2	90.54	87.33	86.23	2.2493
3	91.08	89.54	87.08	2.2286
4	90.35	88.57	85.98	2.3152
5	89.79	85.14	86.09	2.3265
Average	90.62	87.89	86.72	2.2583

In the realm of deep learning, the loss function plays a pivotal role in the training and optimization of the model. It quantifies the difference between the model's predictions and the actual labels. By refining the loss function, we can better align the model's outputs with the true labels in the training dataset, thereby improving the model's capacity to generalize. In this experiment, the evaluation metrics of accuracy and sensitivity, as illustrated in Figure 4, demonstrate favorable results.

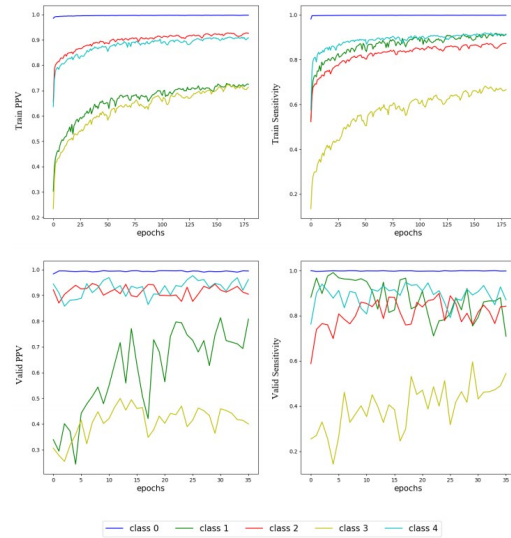


Figure 4. Accuracy and sensitivity trends during the training and validation phases of the multi-class focal loss mode

Furthermore, when compared to models like U-Net, multimodal fusion U-Net, and dual-channel U-Net, the Res-UNet model shows enhanced performance in the task of brain tumor segmentation. Detailed metrics can be found in

Table III, while the segmentation outcomes of selected models are depicted in Figure 5.

TABLE III. PERFORMANCE METRICS OF DIFFERENT MODELS

Model	DSC	PPV	Sensitivity
U-Net[1]	0.61	0.62	0.63
Multimodal Fusion U-Net	0.75	0.74	0.86
multimodal Fusion U-Net[13]	0.84	0.85	0.81
Res-UNet[14]	0.86	0.86	0.87

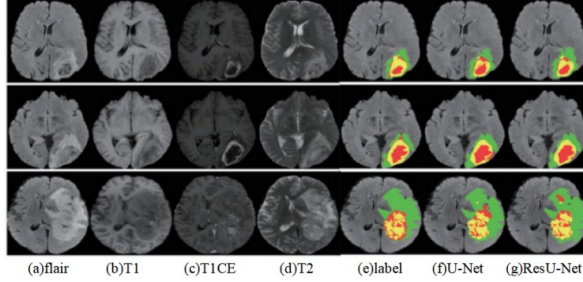


Figure 5. Segmentation results of selected models.

To further validate the effectiveness of multimodal fusion, an ablation study was conducted. The results demonstrate that multimodal image fusion significantly enhances model performance. When trained solely on T1 images, the performance is suboptimal. As additional modalities are incorporated, the model's performance gradually improves, reaching the optimal segmentation results when all four modalities are included, as shown in Table IV.

TABLE IV. EFFECTIVENESS EXPERIMENT OF MULTIMODAL IMAGE FUSION

Modality	DSC	Sensitivity	PPV
T1	0.80	0.77	0.82
T1+T2	0.82	0.82	0.83
T1+T2+T1CE	0.84	0.84	0.84
T1+T2+T1CE+FLAIR	0.85	0.85	0.85

Finally, by comparing the unweighted fusion and weighted fusion methods for multimodal data, the weighted fusion method demonstrates superior segmentation accuracy, as shown in Table V.

TABLE V. COMPARATIVE EXPERIMENT ON THE EFFECTIVENESS OF WEIGHTED FUSION METHODS FOR MULTIMODAL DATA

Fusion Method	DSC
Unweighted Fusion	85.23%
Weighted Fusion	86.75%

V. CONCLUSION

In conclusion, the Res-UNet model developed in this study, which utilizes multimodal fusion, has demonstrated encouraging results in the segmentation of brain tumors from MR images. Through appropriate data preprocessing, model improvements, and training optimization, the model can accurately segment brain tumors, providing a valuable tool for clinical diagnosis. Looking ahead, additional research could be pursued on more intricate clinical cases to enhance the model's adaptability and precision for various tumor types and stages of disease. Additionally, exploring

the deep integration of multimodal data with other clinical information, such as genetic data and patient medical history, could lead to the development of a more comprehensive and precise brain tumor diagnostic model.

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